Supplementary material: the sketched landing method for large-scale optimization under orthogonality constraints *

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Convergence analysis

Recall that we consider the optimization problem

$$\begin{aligned} \underset{X\in\mathbb{R}^{n\times p}}{\text{minimize}} \quad & f(X) \coloneqq \mathbb{E}_{\xi\sim\mathcal{D}}[f_\xi(X)] \\ \text{subject to} \quad & X^\top X = I_p, \end{aligned} \tag{P}$$

on the standard Stiefel manifold $\mathrm{St}(n,p) = \{X \in \mathbb{R}^{n\times p} : X^\top X = I_p\}$, with constraint map $h(X) = X^\top X - I_p$, infeasibility measure $\mathcal{N}(X) = \frac{1}{2}\|h(X)\|_{\mathrm{F}}^2$, and safe region

$$\mathcal{M}^r \;=\; \big\{X \in \mathbb{R}^{n\times p} :\; \|h(X)\|_{\mathrm{F}} \le r\big\}, \qquad r \in (0,1).$$

The (deterministic) landing field on $\mathrm{St}(n,p)$ reads

$$\Lambda(X) \;=\; \omega_{\mathrm{T}}\,\mathrm{grad} f(X) \;+\; \omega_{\mathrm{N}}\,\nabla\mathcal{N}(X), \qquad \begin{aligned} \mathrm{grad} f(X) &= 2\,\mathrm{skew}\big(\nabla f(X) X^\top\big) X, \\ \nabla\mathcal{N}(X) &= 2X\big(X^\top X - I_p\big), \end{aligned}$$

where $\mathrm{skew}(A) = \frac{1}{2}(A - A^\top)$ and $\omega_{\mathrm{T}}, \omega_{\mathrm{N}} > 0$. Writing $B_\zeta = S_\zeta S_\zeta^\top$ for the sketch estimate of the identity, the sketched landing field is

$$\begin{aligned} \Lambda_{\xi,\zeta,\zeta'}(X) &= \omega_{\mathrm{T}} d_{\mathrm{T}}^{\mathrm{sketch}}(X) + \omega_{\mathrm{N}} d_{\mathrm{N}}^{\mathrm{sketch}}(X) \\ &= 2\omega_{\mathrm{T}}\,\mathrm{skew}\big(\nabla f_\xi(X) X^\top B_\zeta\big) B_{\zeta'} X \;+\; 2\omega_{\mathrm{N}}\, B_{\zeta'} X \big(X^\top B_\zeta X - I_p\big). \end{aligned} \tag{0.1}$$

We now make the assumptions precise. Constants below are understood as suprema over the safe region $\mathcal{M}^r$.

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A1 (Objective and stochastic gradient). *Each f_ξ is L-smooth and $f = \mathbb{E}_{\xi\sim\mathcal{D}}[f_\xi]$, so that f is L-smooth. The stochastic gradient is unbiased, $\mathbb{E}_\xi[\nabla f_\xi(X)] = \nabla f(X)$, and has bounded variance: there is $\sigma_G^2 < \infty$ with $\mathbb{E}_\xi\big[\|\nabla f_\xi(X) - \nabla f(X)\|^2\big] \le \sigma_G^2$ for all $X \in \mathcal{M}^r$.*

A2 (Unbiased sketches with finite moments). *The sketches S_ζ are drawn so that $B_\zeta = S_\zeta S_\zeta^\top$ is an unbiased estimate of the identity, $\mathbb{E}_\zeta[B_\zeta] = I_n$, with finite second moment $p_B := \mathbb{E}_\zeta\|B_\zeta\|_2^2 < \infty$ and finite variance $\sigma_B^2 := \mathbb{E}_\zeta\|B_\zeta - I_n\|^2 < \infty$.*

A3 (Independence). *At each iteration the random variables ξ, ζ, ζ' are mutually independent, and independent across iterations.*

A4 (Safe region). *The initial iterate $X_0 \in \mathcal{M}^r$, and almost surely every iterate, together with the line segment joining two consecutive iterates, remains in $\mathcal{M}^r$. Moreover r is small enough that $\mathcal{M}^r$ is compact and every $X \in \mathcal{M}^r$ has full column rank.*

Remark 0.1. *Assumption 2 is satisfied by both sketch families of Section IV. For the sparse (column-subsampling) sketch, $\|B_\zeta\|_2 = n/k$ deterministically, so $p_B = (n/k)^2$ and $\sigma_B^2 < \infty$; for the dense (scaled Haar-distributed) sketch, the moments are finite by finiteness of the Gaussian moments. Unbiasedness $\mathbb{E}_\zeta[B_\zeta] = I_n$ holds by construction. Assumption 4 is the stochastic analogue of the standard "iterates remain in the safe region" hypothesis of landing analyses; as noted in [1, §2.2].*

The following lemma is the crux of the analysis: it shows that a single sketched step is an *unbiased* estimator of the deterministic landing field, with *bounded variance*. This is exactly what is needed to view the sketched landing as an instance of the stochastic landing iteration of [1] and to invoke [1, Theorem 2.9].

Lemma 0.1 (The sketched landing field is unbiased with bounded variance). *Let Assumptions 1 to 4 hold and fix $X \in \mathcal{M}^r$. Then the sketched landing field (0.1) satisfies*

$$\mathbb{E}_{\xi,\zeta,\zeta'}\big[\Lambda_{\xi,\zeta,\zeta'}(X)\big] = \Lambda(X),$$

and, with $\widetilde{E}(X,\Xi) := \Lambda_{\xi,\zeta,\zeta'}(X) - \Lambda(X)$ and $\Xi = (\xi,\zeta,\zeta')$,

$$\mathbb{E}_{\xi,\zeta,\zeta'}\big[\|\widetilde{E}(X,\Xi)\|^2\big] \;\le\; \gamma^2 \;:=\; \sigma_G^2\,\alpha_G + \sigma_B^2\big(\alpha_B + \omega_{\mathrm{N}}^2\gamma_B\big) \;<\; \infty,$$

where the finite constants specialise [1, Prop. 3.3] to $B = I_n$ (i.e. $\beta_1 = \beta_n = \kappa_B = 1$):

$$\alpha_G = 8\,p_B^2(1+r)^2, \quad \alpha_B = 8(1+r)\,\Delta\,(p_B+1), \quad \gamma_B = 8(1+r)\big[(1+r)\sigma_B^2 + r^2 + (1+r)^3\big],$$

with $\Delta := \sup_{X\in\mathcal{M}^r}\|\nabla f(X)X^\top\|_2^2 < \infty$.

Proof. Unbiasedness. By Assumption 3, ξ, ζ, ζ' are mutually independent, and $\mathrm{skew}(\cdot)$ is linear. For the tangent term of (0.1), factorizing the expectation over the independent blocks ζ' and (ξ,ζ) gives

$$\mathbb{E}\big[\mathrm{skew}(\nabla f_\xi(X)X^\top B_\zeta)\,B_{\zeta'}X\big] = \mathbb{E}\big[\mathrm{skew}(\nabla f_\xi(X)X^\top B_\zeta)\big]\,\mathbb{E}[B_{\zeta'}]\,X = \mathrm{skew}\big(\mathbb{E}[\nabla f_\xi(X)]\,X^\top\mathbb{E}[B_\zeta]\big)X.$$

Using $\mathbb{E}_\xi[\nabla f_\xi(X)] = \nabla f(X)$ (Assumption 1) and $\mathbb{E}_\zeta[B_\zeta] = \mathbb{E}_{\zeta'}[B_{\zeta'}] = I_n$ (Assumption 2), this equals $\mathrm{skew}(\nabla f(X)X^\top)X$. Likewise for the normal term,

$$\mathbb{E}\big[B_{\zeta'}X(X^\top B_\zeta X - I_p)\big] = \mathbb{E}[B_{\zeta'}]\,X\big(X^\top\mathbb{E}[B_\zeta]X - I_p\big) = X\big(X^\top X - I_p\big).$$

Summing with weights $\omega_{\mathrm{T}}, \omega_{\mathrm{N}}$ yields $\mathbb{E}[\Lambda_{\xi,\zeta,\zeta'}(X)] = \Lambda(X)$.

Bounded variance. The field (0.1) is precisely the stochastic generalized Stiefel landing field of [1, Eq. (2)] with the constraint matrix estimated by $B_\zeta = S_\zeta S_\zeta^\top$. Its variance is therefore bounded by [1, Prop. 3.3], which for a general B reads $\mathbb{E}_\Xi[\|\widetilde{E}(X,\Xi)\|^2] \le \sigma_G^2\alpha_G + \sigma_B^2(\alpha_B + \omega_{\mathrm{N}}^2\gamma_B)$. Setting $B = I_n$ (so $\beta_1 = \beta_n = 1$) in [1, Eqs. (74)–(76)] gives the constants above. Each is finite: $\sigma_G^2 < \infty$ by Assumption 1; $p_B, \sigma_B^2 < \infty$ by Assumption 2; and $\Delta < \infty$ because ∇f is continuous on the compact set $\mathcal{M}^r$ (Assumption 4) and $\|X\|_2 \le \sqrt{1+r}$ there. Hence $\gamma^2 < \infty$. □

Theorem 0.2 (Convergence of the sketched landing method)**.** *Let Assumptions 1 to 4 hold, and run the sketched landing Algorithm III.1 (or its variance-reduced version Algorithm V.1) with diminishing step sizes $\alpha_t = \alpha_0(1+t)^{-1/2}$. Then there is a threshold $\bar{\alpha}_0 > 0$ such that, for every $\alpha_0 \le \bar{\alpha}_0$, the iterates satisfy*

$$\inf_{t\le T} \mathbb{E}\big[\|\mathrm{grad} f(X_t)\|_{\mathrm{F}}^2\big] \;\le\; \widetilde{\mathcal{O}}\Big(\frac{1}{\sqrt{T}}\Big), \qquad \inf_{t\le T} \mathbb{E}\big[\|h(X_t)\|_{\mathrm{F}}^2\big] \;\le\; \widetilde{\mathcal{O}}\Big(\frac{1}{\sqrt{T}}\Big).$$

More explicitly, with $\rho > 0$ the relative-ascent parameter of $\Psi := \omega_{\mathrm{T}}\mathrm{grad} f$, $L_{\mathcal{L}}$ the smoothness constant of Fletcher's augmented Lagrangian $\mathcal{L}$, $\mathcal{L}^* = \min_{\mathcal{M}^r}\mathcal{L}$, and γ^2 as in Lemma 0.1,

$$\inf_{t\le T} \mathbb{E}\big[\|h(X_t)\|^2\big] \;\le\; \frac{4\big(\mathcal{L}(X_0) - \mathcal{L}^*\big)}{\rho\,\alpha_0\sqrt{T}} + \frac{2\alpha_0\gamma^2 L_{\mathcal{L}}\big(1+\log(T+1)\big)}{\rho\sqrt{T}},$$

and the same bound holds for $\omega_{\mathrm{T}}^2 \inf_{t\le T}\mathbb{E}[\|\mathrm{grad} f(X_t)\|^2]$.

Proof. We verify that the sketched landing iteration $X_{t+1} = X_t - \alpha_t\Lambda_{\xi,\zeta,\zeta'}(X_t)$ meets every hypothesis of [1, Thm. 2.9], applied on $\mathrm{St}(n,p)$ (i.e., $B = I_n$) with relative ascent direction $\Psi(X) := \omega_{\mathrm{T}}\mathrm{grad} f(X)$, penalty weight $\omega := \omega_{\mathrm{N}}$, and noise $\widetilde{E}(X,\Xi) = \Lambda_{\xi,\zeta,\zeta'}(X) - \Lambda(X)$, so that the iteration takes the form $X_{t+1} = X_t - \alpha_t\big[\Lambda(X_t) + \widetilde{E}(X_t,\Xi_t)\big]$ of [1, Eq. (7)].

Assumption 2.1 (smoothness of f). Immediate from Assumption 1: f is L-smooth.

Relative ascent direction. By [1, Prop. 2.3], $\mathrm{grad} f$ is a relative ascent direction (Definition 2.1) with parameter 1; scaling by $\omega_{\mathrm{T}} > 0$ leaves conditions (i)–(iii) intact and yields $\Psi = \omega_{\mathrm{T}}\mathrm{grad} f$ with some $\rho > 0$.

Assumption 2.2 (constraint smoothness). For $h(X) = X^\top X - I_p$ we have $Dh(X)^* = 2X$. Specialising [1, Prop. 3.1] to $B = I_n$, on $\mathcal{M}^r$ the singular values obey $\sqrt{1-r} \le \sigma_i(X) \le \sqrt{1+r}$, hence $\sigma\big(Dh(X)\big)$ is bounded above and below by positive constants $C_h, \bar{C}_h$, and $\mathcal{N}$ is $L_{\mathcal{N}}$-smooth on $\mathcal{M}^r$ with $L_{\mathcal{N}} = 2\big(r + 2(1+r)\big)$. Full rank on $\mathcal{M}^r$ holds by Assumption 4.

Assumption 2.3 (bounded multiplier differential). The multiplier $\lambda(X)$ of Fletcher's augmented Lagrangian is the smooth solution of the Lyapunov equation of [1, §D.1], well defined since X has full rank on $\mathcal{M}^r$. As a smooth map on the compact set $\mathcal{M}^r$ (Assumption 4) it has bounded differential, $\sup_{X\in\mathcal{M}^r}\|D\lambda(X)\| \le C_\lambda < \infty$.

Assumption 2.4 (bounded relative ascent direction). $\Psi(X) = 2\omega_{\mathrm{T}}\,\mathrm{skew}(\nabla f(X)X^\top)X$ is continuous on the compact set $\mathcal{M}^r$, hence $\sup_{X\in\mathcal{M}^r}\|\Psi(X)\| \le C_\Psi < \infty$.

Assumption 2.5 (unbiased estimator with bounded variance). This is exactly Lemma 0.1: $\mathbb{E}_\Xi[\widetilde{E}(X,\Xi)] = 0$ and $\mathbb{E}_\Xi[\|\widetilde{E}(X,\Xi)\|^2] \le \gamma^2 < \infty$ on $\mathcal{M}^r$.

Safe-region hypothesis. [1, Thm. 2.9] further requires that the segments joining successive iterates lie in $\mathcal{M}^r$ almost surely; this is Assumption 4.

All hypotheses being met, [1, Thm. 2.9] applies with the step size $\alpha_t = \alpha_0(1+t)^{-1/2}$ and any $\alpha_0 \le \bar{\alpha}_0 := \frac{\rho}{2L_{\mathcal{L}}} \min\{1, (\omega_{\mathrm{N}} C_h)^{-2}\}$, giving

$$\inf_{t \le T} \mathbb{E}\big[\|\Psi(X_t)\|^2\big] \le \frac{4(\mathcal{L}(X_0) - \mathcal{L}^*)}{\rho\, \alpha_0 \sqrt{T}} + \frac{2\alpha_0 \gamma^2 L_{\mathcal{L}}(1 + \log(T+1))}{\rho \sqrt{T}},$$

and the identical bound for $\inf_{t \le T} \mathbb{E}[\|h(X_t)\|^2]$. Since $\|\Psi(X)\|^2 = \omega_{\mathrm{T}}^2 \|\mathrm{grad} f(X)\|^2$, dividing by ω_{T}^2 gives the stated bound on the Riemannian gradient. Both right-hand sides are $\widetilde{\mathcal{O}}(1/\sqrt{T})$. □

Remark 0.2 (SVRG variant)**.** *The conclusion of Theorem 0.2 also holds for the SVRG sketched landing (Algorithm V.1). Conditionally on the reference point Y, its directions $d^{\mathrm{svrg}}(X_t) = d^{\mathrm{landing}}(Y) - d^{\mathrm{sketch}}(Y) + d^{\mathrm{sketch}}(X_t)$ are still unbiased estimators of $\Lambda(X_t)$, because $\mathbb{E}[d^{\mathrm{sketch}}(Y)] = d^{\mathrm{landing}}(Y) = \Lambda(Y)$ and $\mathbb{E}[d^{\mathrm{sketch}}(X_t)] = \Lambda(X_t)$ by Lemma 0.1; the control variate has zero conditional mean. The variance stays bounded on $\mathcal{M}^r$ by the same argument (each term is bounded-variance and $Y \in \mathcal{M}^r$ by Assumption 4), so Assumption 2.5 holds and [1, Thm. 2.9] applies verbatim.*

Additional details about the computational complexity of the sketched landing with sparse sketches

This section provides additional details about probabilities, expectations and the computational complexity of the sketched landing with sparse random sketches. In particular, we give a more detailed explanation of the complexity $\mathcal{O}(pk^2)$ flops for computing the landing direction. We start by showing two results regarding the sparse random sketches.

Lemma 0.3. *Assume $n \ge k > 0$ and let ζ and ζ' be two lists of $k > 0$ distinct integers uniformly sampled between 1 and n. Then*

$$\mathbb{E}_{\zeta,\zeta'}[|\zeta \cap \zeta'|] = \frac{k^2}{n}.$$

Proof. For $i = 1, 2, ..., k$, let X_i be indicator variable such that

$$X_i = \begin{cases} 1 \text{ if } \zeta_i \in \zeta', \\ 0 \text{ otherwise.} \end{cases}$$

This yields

$$\begin{aligned} \mathbb{E}_{\zeta,\zeta'}[|\zeta \cap \zeta'|] &= \mathbb{E}[\sum_{i=1}^{k} X_i] \\ &= \sum_{i=1}^{k} \mathbb{E}[X_i] \\ &= \sum_{i=1}^{k} \mathbb{P}[X_i = 1] \\ &= \sum_{i=1}^{k} \frac{k}{n} \\ &= \frac{k^2}{n}. \end{aligned}$$

□

Lemma 0.4. *Assume $n \geq k > 0$ and let ζ and ζ' be two lists of $k > 0$ distinct integers uniformly sampled between 1 and n. Then*

$$\mathbb{P}_{\zeta,\zeta'}[\zeta \cap \zeta' = \emptyset] = \frac{(n-k)!^2}{n!(n-2k)!}.$$

Proof. Let the list ζ' be already sampled. Let us compute the probability of sampling a list ζ that is disjoint from the list ζ'. For the first element of ζ, we have $\mathbb{P}_{\zeta_1}[\zeta_1 \notin \zeta'] = \frac{n-k}{n}$. Then, for the second element, we have $\mathbb{P}_{\zeta_2}[\zeta_2 \notin \zeta' \mid \zeta_1 \notin \zeta] = \frac{n-k-1}{n-1}$, etc. Therefore,

$$\begin{aligned}
\mathbb{P}_{\zeta,\zeta'}[\zeta \cap \zeta' = \emptyset] &= \prod_{i=1}^{k} \mathbb{P}_{\zeta_i}[\zeta_i \notin \zeta' \mid \{\zeta_t\}_{t=1}^{i-1} \notin \zeta] \\
&= \prod_{i=1}^{k} \frac{n-k-(i-1)}{n-(i-1)} \\
&= \frac{n-k}{n} \cdot \frac{n-k-1}{n-1} \cdot \cdots \cdot \frac{n-2k+1}{n-k+1} \\
&= \frac{(n-k)!^2}{n!(n-2k)!}.
\end{aligned}$$

□

If $k \ll n$, the result of Lemma 0.4 can be made more practical. Using the approximation

$$\frac{n-k-(i-1)}{n-(i-1)} = 1 - \frac{k}{n-(i-1)} \approx 1 - \frac{k}{n},$$

it follows that $\mathbb{P}_{\zeta,\zeta'}[\zeta \cap \zeta' = \emptyset] \approx \left(1 - \frac{k}{n}\right)^k \approx e^{\frac{-k^2}{n}}$. Thus, if $(n,k) = (1000, 10)$, we have $\zeta \cap \zeta' = \emptyset$ with 90% chance.

Let us now discuss some consequences of Lemmas 0.3 and 0.4 in the complexity analysis of the sketched landing method. The expression of the sketched tangent component of the landing direction is given by

$$d_{\mathrm{T}}^{\mathrm{sketch}}(X) = -\left[\nabla f_\xi(X)\left(\left(S_\zeta^\top X\right)^\top (S_\zeta^\top S_{\zeta'})\right)\right]\left(S_{\zeta'}^\top X\right) + S_\zeta\left[\left(\left(S_\zeta^\top X\right)\left(\nabla f_\xi(X)^\top S_{\zeta'}\right)\right)\left(S_{\zeta'}^\top X\right)\right].$$

First, in view of Lemma 0.4, $S_\zeta^\top S_{\zeta'} = 0$ with probability $\frac{(n-k)!^2}{n!(n-2k)!}$, the computation reduces to

$$d_{\mathrm{T}}^{\mathrm{sketch}}(X) = S_\zeta\left[\left(\left(S_\zeta^\top X\right)\left(\nabla f_\xi(X)^\top S_{\zeta'}\right)\right)\left(S_{\zeta'}^\top X\right)\right].$$

In this case, it is simple to verify that $d_{\mathrm{T}}^{\mathrm{sketch}}(X)$ is computed with $\mathcal{O}(pk^2)$ flops using the association suggested by the parenthesis. We recall that X and $\nabla f_\xi(X)$ are $n \times p$ matrices.

If $S_\zeta^\top S_{\zeta'} \neq 0$, it is important to use the decomposition $S_\zeta^\top S_{\zeta'} = P_{\mathrm{l}} D P_{\mathrm{r}}$ where D is diagonal with $|\zeta \cap \zeta'|$ positive diagonal entries and $P_{\mathrm{l}}, P_{\mathrm{r}}$ are permutation matrices. Then, it follows that

$$\left[\nabla f_\xi(X)\left(\left(S_\zeta^\top X\right)^\top (S_\zeta^\top S_{\zeta'})\right)\right]\left(S_{\zeta'}^\top X\right) = \left[\nabla f_\xi(X)\left(\left(S_\zeta^\top X\right)^\top (P_{\mathrm{l}}\sqrt{D})\right)\right]\left(\sqrt{D}P_{\mathrm{r}}\left(S_{\zeta'}^\top X\right)\right).$$

By Lemma 0.3, D has only $\frac{k^2}{n}$ nonzero diagonal entries in expectation. Therefore, the product $\left(S_\zeta^\top X\right)^\top (P_1\sqrt{D})$ has only $\frac{k^2}{n}$ nonzero columns in expectation. The same holds for $\sqrt{D}P_\mathrm{r}\left(S_{\zeta'}^\top X\right)$ with the rows. The matrix $\nabla f_\xi(X)$ is $n \times p$ and the product $\nabla f_\xi(X)\left(\left(S_\zeta^\top X\right)^\top (P_1\sqrt{D})\right)$ costs $\mathcal{O}(np\frac{k^2}{n}) = \mathcal{O}(pk^2)$ flops in expectation. Finally the multiplication by the right by $\sqrt{D}P_\mathrm{r}\left(S_{\zeta'}^\top X\right)$ also costs $\mathcal{O}(pk^2)$ flops in expectation.

For the sketched normal component of the landing term, we have

$$d_\mathrm{N}^\mathrm{sketch}(X) = -2S_{\zeta'}\left[\left((S_{\zeta'}^\top X)(S_\zeta^\top X)^\top\right)(S_\zeta^\top X)\right] + 2S_{\zeta'}(S_{\zeta'}^\top X).$$

In this case, the complexity of $\mathcal{O}(pk^2)$ flops follows without any particular trick. We have thus shown that computing the landing direction from $X, \nabla f_\xi(X), S_\zeta$ and $S_{\zeta'}$ costs $\mathcal{O}(pk^2)$ flops in expectation.

References

[1] Simon Vary, Pierre Ablin, Bin Gao, and Pierre-Antoine Absil. Optimization without Retraction on the Random Generalized Stiefel Manifold. In *Proceedings of the 41st International Conference on Machine Learning*, pages 49226–49248. PMLR, July 2024.